

Generative AI – Complete Detailed Guide

1. What is Generative AI?

Generative AI is a branch of artificial intelligence that can create new content such as text, images, audio, video, and code.

Unlike traditional AI that only analyzes or classifies data, Generative AI produces new outputs by learning patterns from large datasets.

Examples include ChatGPT for text, image generators, music generators, and code assistants.

2. Core Concepts Behind Generative AI

Generative AI works using deep learning and neural networks. Key concepts include:

- Neural Networks – layered models inspired by the human brain
- Transformers – modern architecture used in large language models
- Tokens – small text units models process
- Embeddings – vector representations of meaning
- Training Data – large datasets used to learn patterns

3. Transformers Architecture

Transformers are the foundation of most modern GenAI systems. They use attention mechanisms to understand context.

Main components:

- Encoder and Decoder blocks
- Self-Attention mechanism
- Multi-Head Attention
- Positional Encoding

Transformers allow models to understand relationships between words across long context windows.

4. Large Language Models (LLMs)

LLMs are trained on massive text corpora to generate and understand language.

Examples: GPT models, Claude, Gemini, LLaMA.

Capabilities:

- Question answering
- Summarization
- Coding
- Translation
- Conversation

Limitations:

- Hallucinations
- Bias
- Context limits

5. Training Process

Training happens in stages:

- Data collection and cleaning
- Tokenization
- Pretraining on large datasets
- Fine-tuning on specific tasks
- Reinforcement Learning with Human Feedback (RLHF)

Training requires high compute and GPUs.

6. Prompt Engineering

Prompt engineering is the skill of writing effective instructions for AI.

Best practices:

- Be specific
- Provide examples
- Define format
- Use step-by-step instructions
- Constrain output

Types:

- Zero-shot
- Few-shot
- Chain-of-thought prompting

7. Retrieval Augmented Generation (RAG)

RAG combines LLMs with external knowledge sources.

Steps:

- Convert documents into embeddings
- Store in vector database
- Retrieve relevant chunks
- Feed to model with prompt

Benefits:

- More accurate answers
- Domain knowledge
- Reduced hallucinations

8. Vector Databases

Vector databases store embeddings for similarity search.

Popular options:

- Pinecone
- FAISS
- Chroma
- Weaviate

Used in semantic search and RAG pipelines.

9. Generative AI Modalities

Generative AI is not only text.

Types:

- Text generation
- Image generation
- Audio generation
- Video generation
- Code generation
- 3D generation

10. Image Generation Models

Models like diffusion models generate images from noise.

Examples:

- Stable Diffusion
- DALL·E
- Midjourney

They use text prompts to guide image creation.

11. Code Generation

AI coding tools generate and explain code.

Examples:

- GitHub Copilot
- ChatGPT
- Cursor

Use cases:

- Autocomplete
- Refactoring
- Debugging
- Architecture help

12. AI Agents

AI agents are systems that can plan, act, and use tools.

Features:

- Goal driven
- Tool usage
- Memory
- Multi-step reasoning

Frameworks:

- LangChain
- AutoGen

- CrewAI

13. Evaluation of GenAI Systems

Evaluation methods:

- Accuracy
- Faithfulness
- Relevance
- BLEU / ROUGE scores
- Human evaluation
- Task success rate

14. Risks and Limitations

Challenges include:

- Hallucinations
- Bias
- Security risks
- Data leakage
- Overreliance

Mitigation requires guardrails and validation.

15. Ethical Considerations

Important topics:

- Responsible AI use
- Transparency
- Fairness
- Privacy
- Consent
- Content authenticity

16. Real World Applications

Applications include:

- Chatbots
- Medical assistants
- Education tutors
- Legal drafting
- Creative writing
- Software development

17. GenAI Tech Stack

Typical stack:

- Frontend app
- Backend API
- LLM provider
- Vector database
- Embedding model
- Retrieval layer
- Monitoring

18. Learning Roadmap

Step path:

- Learn Python
- Understand ML basics
- Study Transformers
- Practice Prompt Engineering
- Build RAG projects
- Create AI apps
- Study evaluation and safety

19. Future of Generative AI

Future trends:

- Multimodal models
- Longer context windows

- Realtime agents
- Personal AI assistants
- Domain-specific models

20. Practice Projects

Project ideas:

- PDF chat assistant
- AI tutor
- Document summarizer
- Code reviewer
- Medical Q&A; bot
- Multilingual chatbot